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IndiaKart E-Commerce Project

Phase 3: KPI Calculations

Kindly view the Python Notebook file at this link: Python Notebook file link

KPI Results and Interpretations

1. Gross Merchandise Value (GMV)

Definition / formula	Sum of final_amount across all orders, regardless of status
Calculated result	₹3,147,385,545
Benchmark / target	None defined
Status	Informational

Interpretation: GMV measures the total value of orders placed over the 24-month period, before cancellations, returns or undelivered orders are removed. It is the starting point for the revenue funnel.

Implications and areas to investigate: GMV alone overstates earned revenue. It should be read alongside Net Revenue to see how much value is lost between order placement and delivery.

2. Net Revenue

Definition / formula	Sum of final_amount where order status = Delivered
Calculated result	₹2,053,858,612 (32,499 delivered orders)
Benchmark / target	None defined
Status	Informational

Interpretation: Delivered orders account for 65.3% of GMV. The remaining 34.7% (₹1,093,526,933, a derived difference) sits in Cancelled (5,996), Returned (3,933), Shipped (5,071) and Processing (2,501) orders.

Implications and areas to investigate: Shipped and Processing orders may still convert to delivered revenue, so Net Revenue is a point-in-time figure. The value split across the non-delivered statuses was not calculated and would show where the largest leakage lies.

3. Average Order Value (AOV)

Definition / formula	Net Revenue ÷ count of delivered orders
Calculated result	₹63,198 per delivered order
Benchmark / target	None defined
Status	Informational

Interpretation: A typical delivered order is worth about ₹63,000, a high-ticket value consistent with Electronics generating most category revenue.

Implications and areas to investigate: Because each order carries high value, a single cancellation or return has a larger revenue effect than in a low-ticket business. AOV by category and segment would be a useful follow-up.

4. Cancellation Rate

Definition / formula	(Cancelled orders ÷ total orders) × 100
Calculated result	11.99% (5,996 of 50,000 orders)
Benchmark / target	Below 10%
Status	Does Not Meet Target

Interpretation: Roughly one in eight orders is cancelled, which is 1.99 percentage points above the ceiling.

Implications and areas to investigate: The analysis does not show why orders are cancelled. Cancellations should be broken down by category, payment method, order stage and time period, and compared with payment failures to test whether the two are linked.

5. Return Rate

Definition / formula	(Return records ÷ delivered orders) × 100
Calculated result	30.77% (10,000 return records ÷ 32,499 delivered orders)
Benchmark / target	Below 8%
Status	Does Not Meet Target

Interpretation: The result is nearly four times the target. No duplicate or unmatched return order IDs were found, but only 3,933 orders have the status "Returned", fewer than the 10,000 return records. The numerator therefore includes returns on orders that are not recorded as Returned, while the denominator counts only Delivered orders.

Implications and areas to investigate: First, reconcile the definition (which order statuses the return records belong to) and recalculate consistently. Then analyse returns by category and return reason. Until then, the figure signals a likely problem, but its size is uncertain.

6. Customer Lifetime Value (CLV)

Definition / formula	Average total_spent per customer segment
Calculated result	Premium ₹532,735 vs Regular ₹361,871: ratio 1.47x
Benchmark / target	Premium about 3x Regular
Status	Does Not Meet Target

Interpretation: This is average recorded lifetime spend, not a predictive CLV. Premium customers spend the most, but only 1.47 times the Regular segment against the expected 3x. Table 1 shows all segments.

Implications and areas to investigate: The weak separation suggests that segment definitions or the recorded total_spent values should be reviewed, or that the 3x benchmark does not fit current behaviour. The Inactive segment (₹32,249) is a clear re-engagement candidate. Customer counts per segment were not reported.

Table 1: Average recorded lifetime spend by customer segment

Segment	Average total_spent (₹)	Index vs Regular (derived)
Premium	₹532,735	1.47x
Regular	₹361,871	1.00x
New	₹274,866	0.76x
Budget	₹186,179	0.51x
Inactive	₹32,249	0.09x

7. Month-over-Month (MoM) Growth

Definition / formula	((This month GMV - last month GMV) ÷ last month GMV) × 100
Calculated result	Latest month (June 2025): -23.74%. Average across months: 14.92%
Benchmark / target	None defined
Status	Informational

Interpretation: June 2025 GMV was ₹103,603,348 against ₹135,849,393 in May 2025. However, the dataset ends on 23 June 2025, so June 2025 is a partial month, and the decline is not a like-for-like comparison. The 14.92% average is inflated by +370.77% in July 2023, because June 2023 was also a partial month (data starts 24 June 2023, GMV ₹27,384,849).

Implications and areas to investigate: Excluding both partial months at each end, the 22 months from August 2023 to May 2025 average 0.51% growth, with swings such as -17.26% in February 2025 and +19.61% in March 2025. This points to a flat, volatile trend rather than sustained growth. Restate growth on complete months, accounting for February's shorter length.

8. Top Category Revenue Share

Definition / formula	Category revenue ÷ total revenue × 100 (order_items.total_price)
Calculated result	Electronics: 57.41% (₹1,642,422,813)
Benchmark / target	Concentration risk if one category exceeds 60%
Status	Meets Target

Interpretation: Electronics is the dominant category, followed by Sports & Fitness (17.38%) and Home & Kitchen (8.54%). The remaining seven categories each contribute 5.18% or less.

Implications and areas to investigate: The result is within the threshold by 2.59 percentage points, so concentration remains a live risk rather than resolved. Electronics performance (returns, stock, payment failures) will have an outsized effect on total results and should be monitored closely.

9. Payment Failure Rate

Definition / formula	(Failed payments ÷ total payments) × 100
Calculated result	3.50% (1,748 of 50,000 payments)
Benchmark / target	Below 2%
Status	Does Not Meet Target

Interpretation: The failure rate is 1.5 percentage points above target, so about 1 in 29 payment attempts fails.

Implications and areas to investigate. Failed payments represent potential revenue leakage. The payments table holds gateway, bank, and payment method fields, so failures can be broken down by these to find whether the problem is concentrated. A link to cancellations needs to be tested.

10. Inventory Fill Rate

Definition / formula	(In Stock SKUs ÷ total SKUs) × 100
Calculated result	92.20% (922 of 1,000 SKUs)
Benchmark / target	Above 95%
Status	Does Not Meet Target

Interpretation: The shortfall is 2.8 percentage points. Of the 78 SKUs not classed as In Stock, 76 are Low Stock and only 2 are Out of Stock.

Implications and areas to investigate: The gap is driven mainly by low stock rather than stock-outs, so timely replenishment could close it. Low-stock SKUs should be reviewed against reorder levels, starting with high-selling Electronics items. Whether Low Stock should count against the fill rate is a definition point to confirm.

Consolidated KPI Summary

Status is assigned only where a benchmark is defined. Informational KPIs have no pass/fail label.

KPI	Calculated Result	Benchmark / Target	Status	Key Finding
1. GMV	₹3,147,385,545	None defined	**Informational**	Total order value over 24 months, before any removals
2. Net Revenue	₹2,053,858,612	None defined	**Informational**	65.3% of GMV converted to delivered revenue
3. AOV	₹63,198	None defined	**Informational**	High ticket value per delivered order
4. Cancellation Rate	11.99%	Below 10%	**Does Not Meet Target**	5,996 orders cancelled; 1.99 pts over ceiling
5. Return Rate	30.77%	Below 8%	**Does Not Meet Target**	Nearly 4x target; definition needs reconciling
6. CLV (Premium / Regular)	1.47x	About 3x	**Does Not Meet Target**	Premium ₹532,735 vs Regular ₹361,871
7. MoM Growth (latest)	-23.74%	None defined	**Informational**	June 2025 is a part-month; trend broadly flat
8. Top Category Share	Electronics 57.41%	Risk if above 60%	**Meets Target**	Close to threshold; heavy reliance on Electronics
9. Payment Failure Rate	3.50%	Below 2%	**Does Not Meet Target**	1,748 failed payments of 50,000
10. Inventory Fill Rate	92.20%	Above 95%	**Does Not Meet Target**	76 Low Stock and 2 Out of Stock SKUs

Key Business Findings

(i) **Only about two-thirds of order value becomes delivered revenue.** Net Revenue is ₹2.05 billion against GMV of ₹3.15 billion (65.3%), with 11.99% of orders cancelled and a further 3,933 orders marked Returned.

(ii) **Returns appear to be the most serious operational gap, but the measure needs validation.** The 30.77% rate is far above the 8% target, yet it mixes return records with a delivered-only base.

(iii) **Customer value is less differentiated than expected.** Premium customers spend 1.47x Regular rather than about 3x, and Inactive customers average only ₹32,249.

(iv) **Revenue depends heavily on one category.** Electronics contributes 57.41%, just under the 60% risk threshold.

(v) **Payment and stock issues both miss target.** Payment failures are 3.50% (target below 2%) and the fill rate is 92.20% (target above 95%), the latter driven by Low Stock SKUs.